



**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2018
YEAR SIX
H2 BIOLOGY (9744)**

Suggested Answers

Question 1

(a)(i)

- Centrioles

(a)(ii)

- Cell division /mitosis /meiosis / prophase (I/II)/ interphase
- There are two pairs/sets of centrioles in the cell

(b)(i)

- Prophase II: 8, Anaphase II: 16

(b)(ii)

- Prophase II: Homologous chromosomes are separated equally into 2 daughter cells/ haploid set of chromosomes is formed at the end of meiosis I/ following cytokinesis
- Anaphase II: Centromeres divide in anaphase II, thus doubling number of centromeres

(b)(iii)

- Chromatin had not condensed into discrete chromosomes with centromeres

(b)(iv)

- X decreases from 5 to 0 arbitrary units
- Homologous chromosomes pair up via synapsis to form bivalents

(b)(v)

- X remains constant at 0 arbitrary units;
- Chiasma formation and crossing over occurs between non-sister chromatids of homologous chromosome

(b)(vi)

- Daughter cells are haploid/ only have 1 set of chromosomes

Question 2

(a)

Key:

A represents the dominant allele for yellow coat colour.
a represents the recessive allele for agouti coat colour.

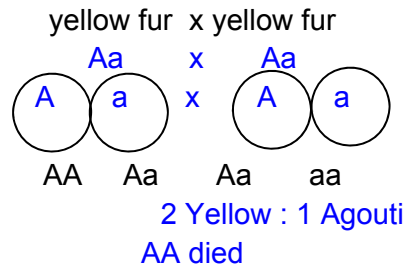
F1 phenotype

F1 genotypes

F1 gametes

F2 genotypes

F2 phenotypic ratio



(b)(i)

Phenotype	Observed (O)	Expected (E)	(O-E) ² / E
Wild-type fur, dark-eyed	12	38	17.789
Wild-type fur, pink-eyed	62	38	15.157
Albino, dark-eyed	78	76	0.0526
Albino, pink-eyed			
Chi Square Value			32.99

(b)(ii)

Since calculated X^2 value 32.95 > critical X^2 value 5.99 at 95% confidence level;

There is a significant difference between observed and expected values, any difference observed are not due to chance;

At 2 degrees of freedom, the calculated value of χ^2 (32.95) lies beyond 9.21. The corresponding probability is $p < 0.01$;

(c)

Key: E represents the dominant allele for dark-eyed.
 e represents the recessive allele for pink-eyed.
 A represents the dominant allele for wild-type fur.
 a represents the recessive allele for albino

F1 Cross:

Phenotype of

parents:

Genotype of
parents:

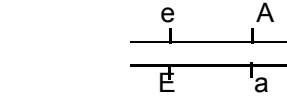
dark-eyed, wild type fur

x

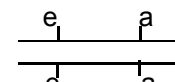
pink-eyed, albino

1 mark

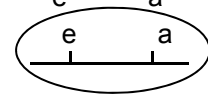
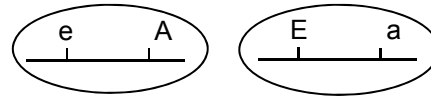
Gametes produced (n):



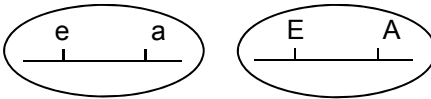
x



x



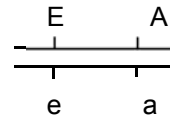
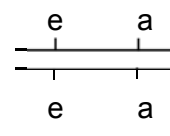
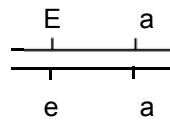
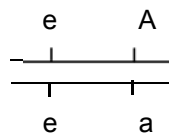
no crossing over



crossing over

1 mark

Genotype of offspring:



1 mark

Phenotype of offspring:

pink-eyed
wild-type furdark-eyed
albinopink-eyed
albinodark-eyed
wild-type fur

1 mark

Question 3

(a)

A

Ligand IGF1 binds to the receptor tyrosine kinase. The receptor cross-phosphorylates each other and becomes activated;

PI 3-kinase binds to the phosphorylated region of receptor tyrosine kinase and becomes activated. Activated PI 3-kinase phosphorylates and activates $PI(4,5)P_2$;

B

Phosphorylated $PI(4,5)P_2$ / $PI(3,4,5)P_3$ serves as a docking site at the cell surface membrane.

PDK1 and Akt binds to the $PI(4,5)P_3$;

Akt is phosphorylated by activated PDK1 and mTOR. Akt becomes activated and dissociates from $PI(3,4,5)P_3$ to phosphorylate Bad / causes Bad to bind to 14-3-3 protein;

(b) (i)

PI 3-kinase is mutated to become locked in the active state / resistant to phosphatases catalysis;

Activity of PI 3-kinase increases / pathway always switched on;

(b) (ii)

Bad protein is mutated such that the conformation of active site involved in binding is changed / unable to bind to apoptosis inhibitory protein;

Bad protein is mutated such that it is able to bind with 14-3-3 protein without phosphorylation / permanently;

Question 4**(a)**

- A codon is three nucleotides/bases on mRNA that specify an amino acid
- It is where the corresponding tRNA anticodon will complementary base-pairs
- signals the start of translation if it is a start codons/AUG
- signals the termination of translation if it is a stop codons/UAA, UAG, UGA

(b)

- Single base pair substitution mutation from guanine to adenine on DNA template strand.
- Results in the substitution/replacement of an amino acid from alanine to valine.

(c)

- Ref. alanine and valine have different R groups
- could result in more extensive hydrophobic interaction between β peptides and thus forming amyloid plaques.

Question 5**(a)**

- ref hydrogen bonds
- Adenine always base pair with Thymine and Cytosine always base pair with Guanine and vice versa / a pyrimidine on one strand always base-pairs with a purine on the other strand
- Thus width between the 2 sugar-phosphate backbones is constant / maintain 2nm
- Complementary DNA strands are antiparallel, thus A-T and G-C pairs would be able to maintain their hydrogen bonds/ antiparallel DNA is also more structurally stable than parallel DNA
- there is hydrophobic interactions between stacked bases

(b)(i)

- Peptidyl transferase
- catalyzes the formation of peptide bonds between amino acids via condensation reaction;

(b)(ii)

- RNA polynucleotide strands / molecules fold into compact 3D shape
- forming an active site with a specific shape
- The shape is maintained by hydrogen bonds between ribonucleotides and interactions with ribosomal protein

(c)

- It prevents the translocation of ribosomes along the mRNA by binding to the empty "E" site, and so only dipeptides are formed

Question 6**(a)**

- Eggs are laid on damp surfaces/ surface of stagnant water
- Larvae hatch from eggs upon being submerged in water
- larvae feed on microorganisms and particulate organic matter, shedding their skins three times to be able to grow from first to fourth **instars**
- When the larva has acquired enough energy and size and is in the fourth instar, metamorphosis is triggered, changing the larva into a pupa
- Pupae do not feed, they just change in form until the body of the adult, flying mosquito is formed and emerged

(b)

- *Aedes aegypti* is a mosquito vector that can spread dengue fever
- In the last 50 year, as climate change to warmer and wetter in more countries, mosquito also thrives in these new places
- therefore where the vector thrives so does the dengue virus

(c)

- ref. human activity provide viable habitats e.g. stagnant water for Aedes to thrive
- In colder latitudes or altitudes, urbanisation / large cities provide a warmer habitat

(d)

- Obtain DNA from both types of mosquito
- Perform PCR to amplify the DNA
- Perform Southern blot to determine if they have similar band/ DNA sequencing

Question 7**(a)(i)**

For (A) and (B):

1 mark for quoting appropriate pair of values with units;

1 mark for describing effect;

E.g.

- A: as conc of CQ increases from 0-25µm, the cells which are infected by H3N2 increase in its viability from 0.9 o.d to 1.5 o.d;
- The higher the conc of CQ, the more virus infected cells are able to survive;
- B: viral RNA copies per reaction decrease from 100000 to 100 as the conc of CQ increase from 0- 25µm;
- The higher the conc of CQ, the fewer viral RNA copies are produced per reaction;

3 max

(a)(ii)

- by preventing fusion of viral and endosomal membranes;
OR
- by preventing conformational change of HA;

(b)

- ref. antigenic drift;
- 8 single-stranded RNA segments;
- spontaneous genetic mutation /mutation during replication;
- lack of proof-reading during RNA replication;
- ref. antigenic shift;
- genetic reassortment of the RNA segments between two strains of viruses;
- novel glycoproteins (spikes) produced on viral envelope / modified neuraminidase and haemagglutinin;
- cannot be recognized by previous antibodies;

4 max

(c)

	Prokaryotic genome	Eukaryotic genome
1. Genome size	Smaller genomes, 0.6 to 10Mb.	Large genomes, being less than 10 Mb – 100,000 Mb
2. Gene length	Shorter gene sequences / more compact genetic organisation	Longer gene sequences / presence of more intergenic spaces
3. Chromosome number	Single chromosome / Haploid	Many chromosomes / Diploid or polyploid
4. Chromosome structure	Circular DNA molecule.	Linear DNA, each contained in a different chromosome.
5. Location	Chromosome found in the nucleoid region of the cytoplasm	Chromosomes are enclosed within a double-membrane bound nucleus
6. Introns	Coding sequence proceeds from start to finish without interruption by introns.	Presence of introns within genes.
7. Repetitive sequences	Few repetitive DNA sequences.	Many highly repetitive DNA sequences
8. Coding and non-coding DNA	Most of DNA are coding sequences (codes for protein, tRNA, or rRNA.	Most of DNA are non-coding.
9. Presence and absence of operons	Two or more genes may be expressed and regulated as a unit (genomes arranged in operons / polycistronic genes)	Absence of operons / monocistronic genes
10. Origins of replication	One	Many
11. Telomeres	Absent	Present
12. Centromere	Absent	Present

Question 8**(a)**

Vaccine needs to be given before girls are sexually active or likely to carry HPV

(b)

The vaccine causes the production of memory cells and antibodies that specifically recognize and bind to antigens from HPV types 16 and 18

Other types of HPV have different antigens and may not be recognized by these memory cells and antibodies

(c)

cancer takes many years to develop

it takes time for young girls to become sexually active

(d)

cancer can be caused by other factors or that the vaccine may not work

(e)

reference to herd immunity;

vaccinating young men would reduce the spread of HPV to females;

Question 9**(a)**

the calls made by cranes are species specific

the greater the similarity in calls, the closer the relationship between different species is likely to be

(b)

The higher the temperature required to separate the hybrid DNA, the higher the percentage of DNA hybridization

As more energy is required to break the hydrogen bonds between base pairs

(c)

- Neanderthals, modern human and chimpanzee are the most closely related
- because the first 20 amino acid sequences for all three organisms are identical
- subsequent relatedness to Neanderthal is in the order of orang-utan, gorilla and monkey
- because orang-utan has 1 amino acid difference from Neanderthals, gorilla 2 differences and monkey 3 differences

Question 10**(a)(i)**

$(8.8\% - 6.9\%) / 20 = 0.10\%$ per minute;

(a)(ii)

1 mark for quoting appropriate pair of values with units;

1 mark for describing appropriate difference (using comparative terms) between solutions A and B;

E.g.

- From 0-20mins, percentage O_2 in solution B increased from 6.9% to 8.95% but percentage O_2 in solution A increased from 6.9% to 8.8%.
- From 0-10min, percentage O_2 in solution B increased more steeply from 6.9% to 8.7% but percentage O_2 in solution A increased from 6.9% to 8.1%.

OR

- From 0-10min, percentage of O_2 in solution B has a higher increase of 1.8% but O_2 increase in solution A increased by only 1.2%.
- From 10-20min, percentage O_2 in solution B increased less steeply from 8.7% to 8.95% but percentage O_2 in solution A increased more steeply from 8.1% to 8.8%.

OR

- From 10-20min, percentage of O_2 in solution B has a lower increase of 0.25% but O_2 increase in solution A has a higher increase of 0.7%.

(a)(iii)

(quote values and describe trend) From 0 – 20 mins, percentage of O_2 in solution C decreased from 6.9% to 6.1%

In the absence of light, no photolysis of water, hence, no oxygen released;

As oxygen is used up in respiration

2 max

(b)(i)

To **phosphorylate** the glucose/fructose-6-phosphate/hexoses

(b)(ii)

- H atoms are accepted by NAD to form reduced NAD
- Electrons are passed to Electron Transport Chain down a progressively lower energy level to provide energy for active transport;
- H^+ ions are actively transported into the intermembrane space via proton pump to generate proton gradient;
- and H^+ ions diffuse via ATP synthase into mitochondrial matrix to generate ATP;